

Public Sentiment and Common Topics on Philippine Agriculture: A Comparative Text Analysis of Reddit Discussions and Online News Articles

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Abstract—This study focuses on public sentiments and common topics of agriculture in the Philippines, specially in news platforms and reddit.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

The main intuition in this study is to provide context into what are the common topics and sentiments towards the topic **agriculture** in the Philippines. While it is common knowledge that agriculture is often **unappreciated** and that it is **unsupported**, this study provides a textual analysis on the conversations that occur in online platforms.

A. Research Questions

- 1) What are the dominant public sentiments toward agriculture in the Philippines across Reddit discussions and news articles?
- 2) What are the most common topics associated with Philippine agriculture in public (Reddit) versus formal (news) discourse?
- 3) How do public sentiments and topics differ between the two sources?
- 4) Is it possible to identify the specific needs or concerns of farmers based on these online discussions?

II. METHODOLOGY

TABLE I
RAW REDDIT DATA SAMPLE

post_title	Sample title about farming
post_url	https://reddit.com/...
comment	Agriculture in the Philippines needs reform.
comment_author	user123
comment_score	12
post_body	Full post discussing rice production and its challenges in the Philippines.
post_score	256
post_author	authorA
created_utc	2025-10-23 08:14:00

TABLE II
ONLINE NEWS PLATFORM DATA SAMPLE

article_title	Government Pushes for Modernized Irrigation Systems to Boost Rice Yields
article_url	https://www.inquirer.net/article/...
article_body	The Department of Agriculture announced new initiatives aimed at improving irrigation efficiency and supporting local farmers struggling with the effects of climate change.
author	Maria Dela Cruz
publisher	Philippine Daily Inquirer
publish_date	2025-10-24 09:30:00
article_category	Agriculture / Economy
article_tags	rice production, irrigation, government policy, farmers

Pipeline

- 1) Data Preprocessing
- 2) Exploratory Data Analysis (EDA)
- 3) Natural Language Processing (NLP)
 - a) Topic Modeling
 - b) Sentiment Analysis

Data Collection & Preprocessing

Reddit data is collected from r/Philippines, while agriculture-related news articles are taken from **Rappler**. Metadata that are gathered from both sources are summarized in their respective tables (See Table I & II)

Exploratory Data Analysis

At this stage, basic exploratory data analysis (EDA) methods are going to provide us a deeper understanding of what the data has to offer. (i) Basic Overview and Structure, (ii) Word Frequency Analysis, (iii) Engagement and Metadata Trends, (iv) Common Keywords, (v) Text Comparison (*Reddit vs News, how do they write text?*)

Natural Language Processing

Natural Language Processing

- a. *Topic Modelling*
- b. *Sentiment Analysis*

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III. RESULTS & DISCUSSION

IV. CONCLUSION

V. POSSIBLE FUTURE WORKS

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